

Clustering of VST-expression trajectories (normalized count data) for genes identified as significantly temporally differentially expressed in heat vs. control)

Pacuta

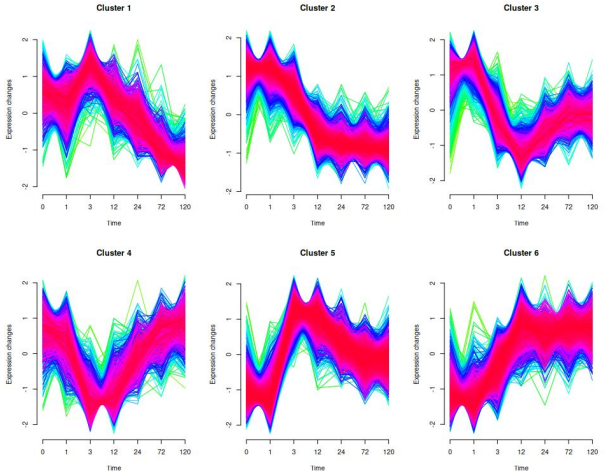
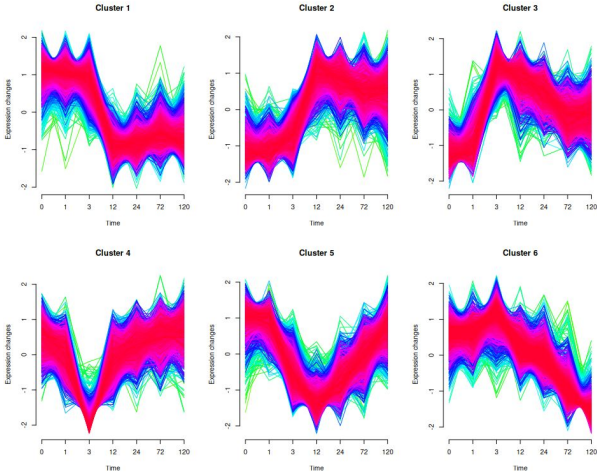
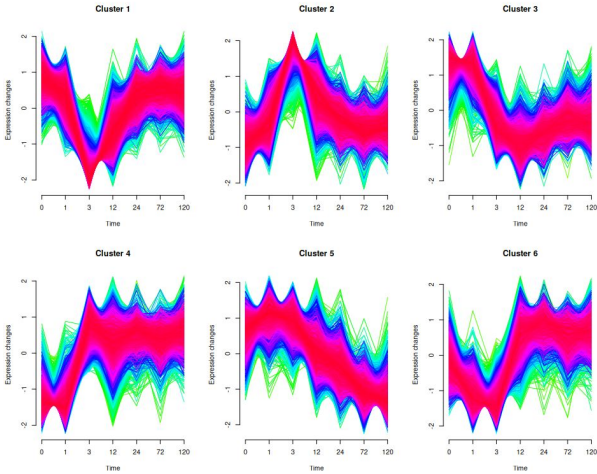
Mcap

Pcomp

cluster	peak_time	trough_time	n_genes
1	0	3	1684
2	3	0	1900
3	0	12	1574
4	3	1	1610
5	1	120	1733
6	24	3	1563

cluster	peak_time	trough_time	n_genes
1	0	12	1270
2	12	0	1096
3	3	0	963
4	120	3	965
5	0	12	1043
6	3	120	1294

cluster	peak_time	trough_time	n_genes
1	3	120	784
2	1	120	1416
3	1	12	1024
4	0	3	675
5	3	1	1260
6	12	1	1074



Color = Cluster Membership

0.0

0.5

1.0

Pacuta

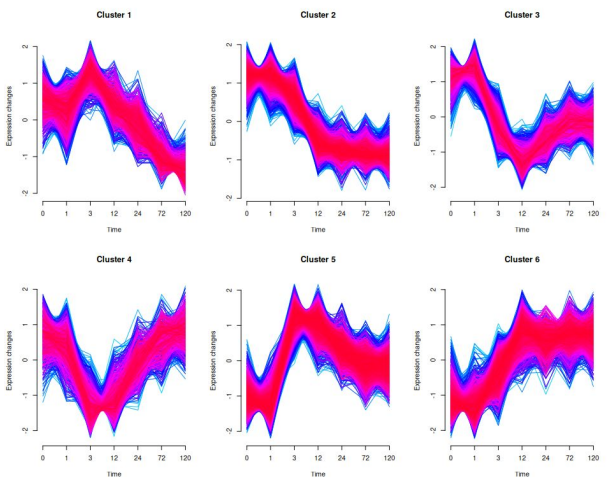
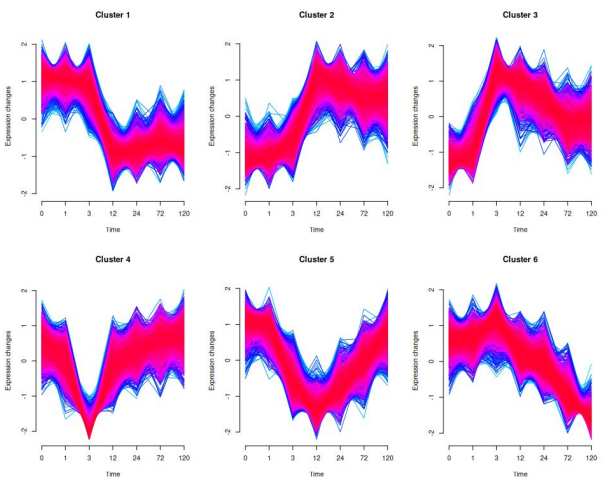
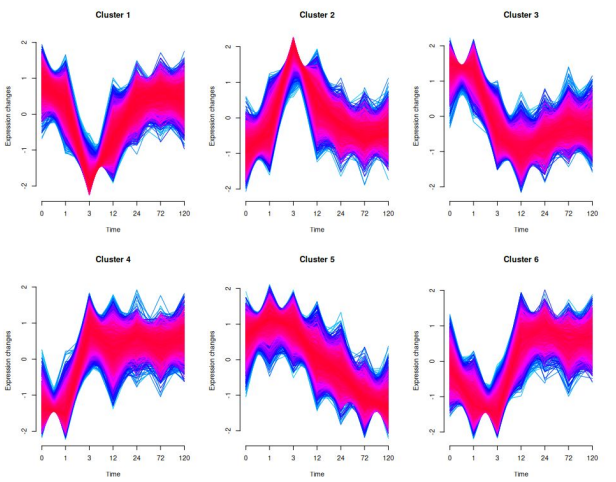
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Filtered for membership > 0.5

Color = Cluster Membership



Pacuta

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Pcomp

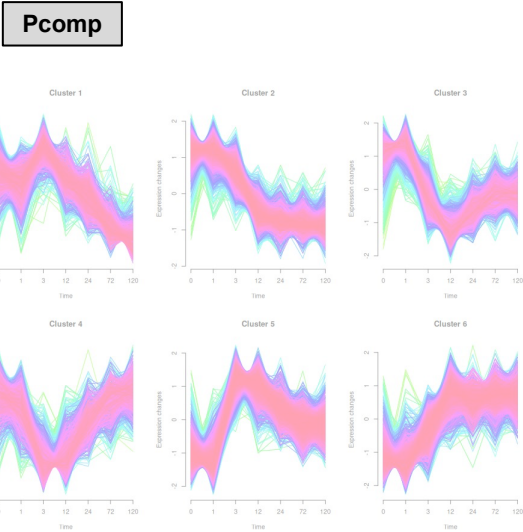
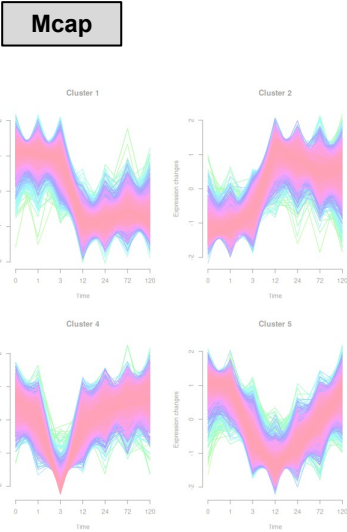
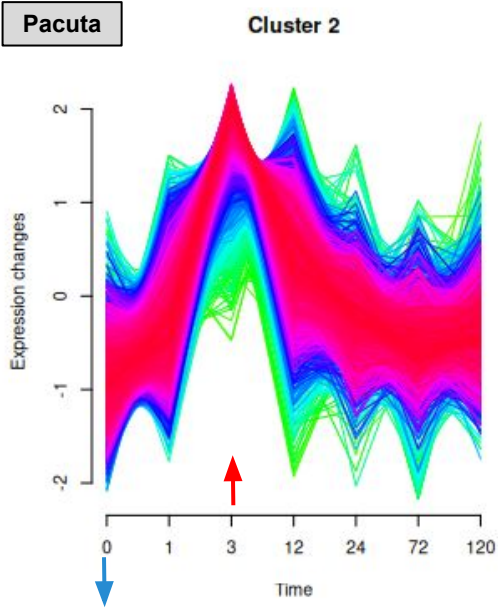
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Pattern	Pacuta	Mcap	Pcomp	Species
Early Peak (3h)	2	-	-	Pacuta
Sustained Up (3h)	4	3	5	All
Sustained Up (12h)	6	2	6	All
Gradual Down	5	6	1	All
Sustained Down (3h)	3	-	3	Pacuta & Pcomp
Sustained Down (12h)	-	1	2	Mcap & Pcomp
Early Dip (3h)	1	4	-	Pacuta & Mcap
U-shaped Dip (12h)	-	5	4	Mcap & Pcomp

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Early Dip (3h)	1	4	-	Pacuta & Mcap
U-shaped Dip (12h)	-	5	4	Mcap & Pcomp

Selected Heat Stress Genes (Manually curated):

- Pacuta
 - ST1C2
 - HSP90
 - HSP70
 - AMPK
 - BI-1
 - Bcl-2
 - BAX
 - CAT



Color = Cluster Membership



Pattern	Pacuta	Mcap	Pcomp	Species
Early Peak (3h)	2	-	-	Pacuta
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Early Dip (3h)	1	4	-	Pacuta & Mcap
U-shaped Dip (12h)	-	5	4	Mcap & Pcomp

Heat Stress Genes (Manually curated):

- NF-KB

-

Mcap

-

Pcomp
- TNFR 27

-

HSP90

-

HSP70
- Pacuta

-

AMPK

-

ALOX12B

-

DNaj

-

BI-1

-

BAX
- HSP90

-

HSP70

-

AMPK

-

Foxo3

-

Cas3, Casp

-

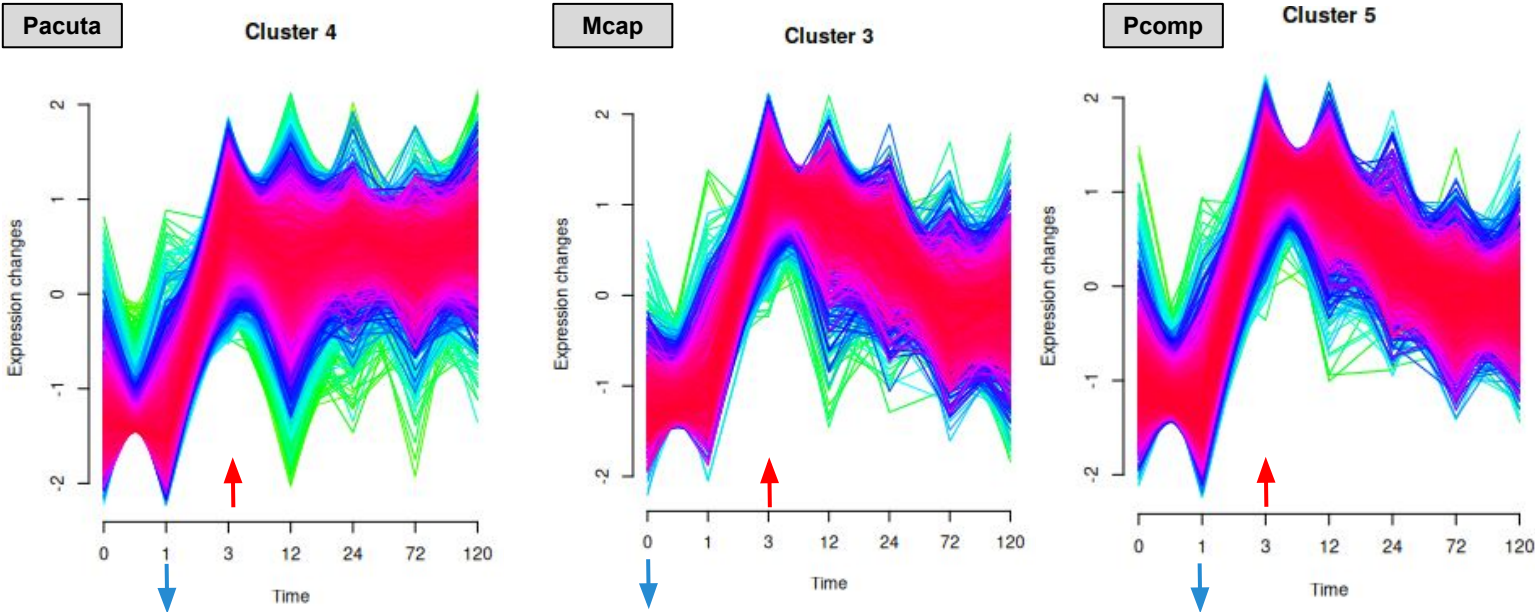
Bcl-2

-

BAX

-

Nrf2



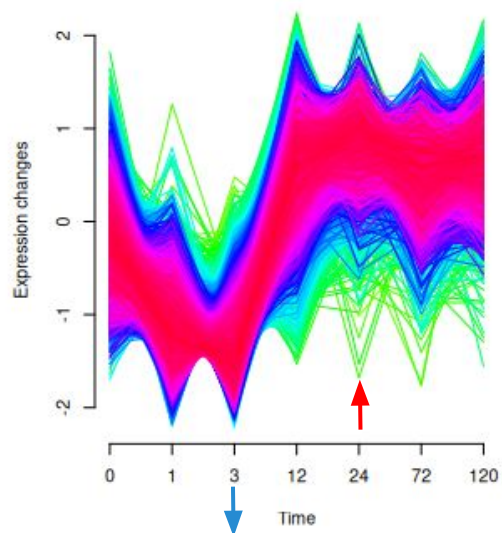
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Early Dip (3h)	1	4	-	Pacuta & Mcap
U-shaped Dip (12h)	-	5	4	Mcap & Pcomp

Heat Stress Genes (Manually curated):

- Pacuta - none
- Mcap
 - ST1C2
 - TNFRSF 27
- Pcomp - none

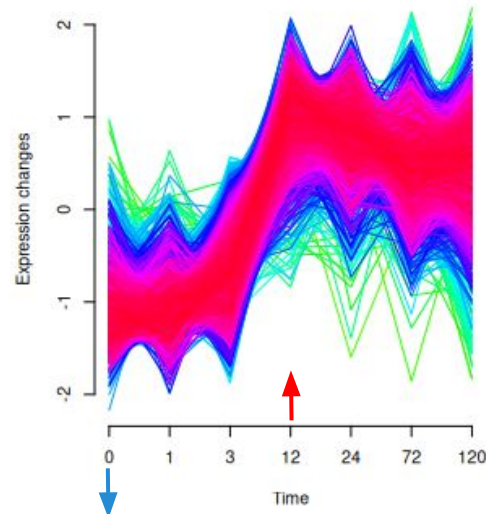
Pacuta

Cluster 6



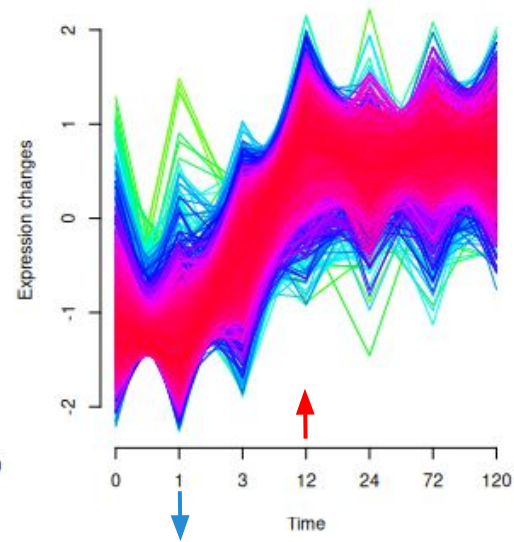
Mcap

Cluster 2



Pcomp

Cluster 6



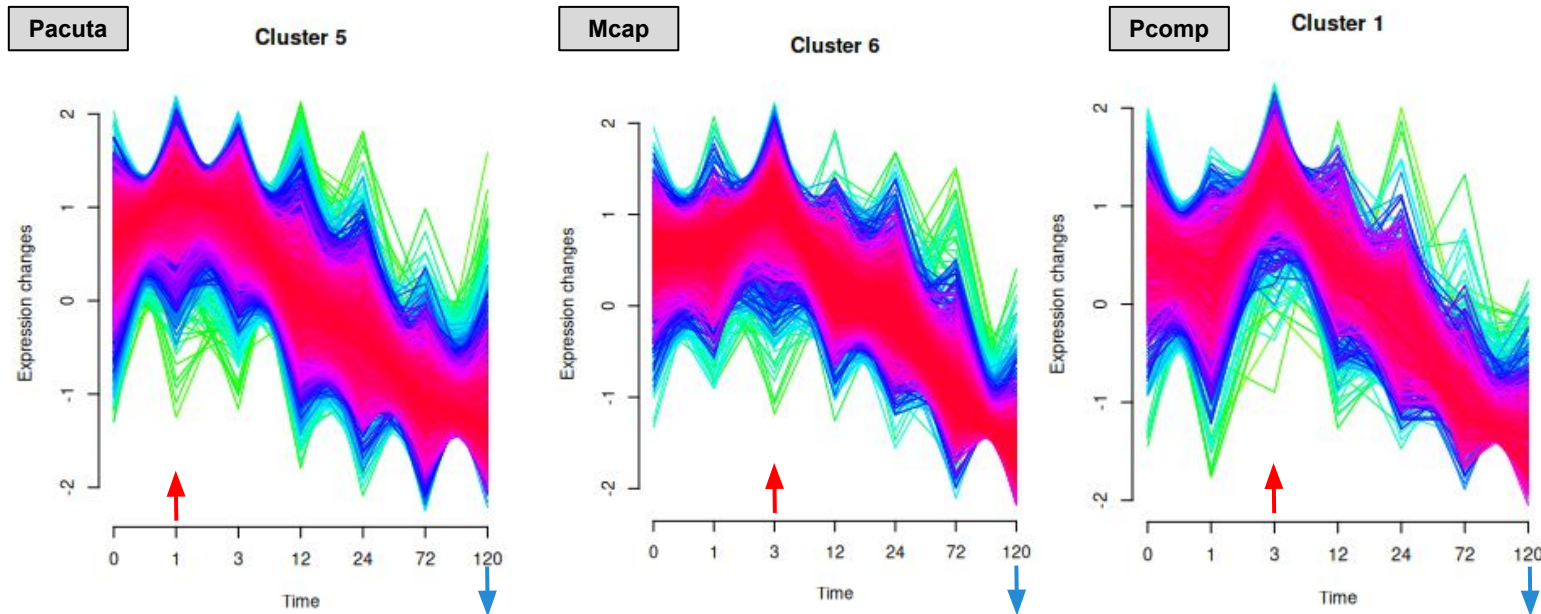
Color = Cluster Membership



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Sustained Down (12h)	-	1	2	Mcap & Pcomp
Early Dip (3h)	1	4	-	Pacuta & Mcap
U-shaped Dip (12h)	-	5	4	Mcap & Pcomp

Heat Stress Genes (Manually curated):

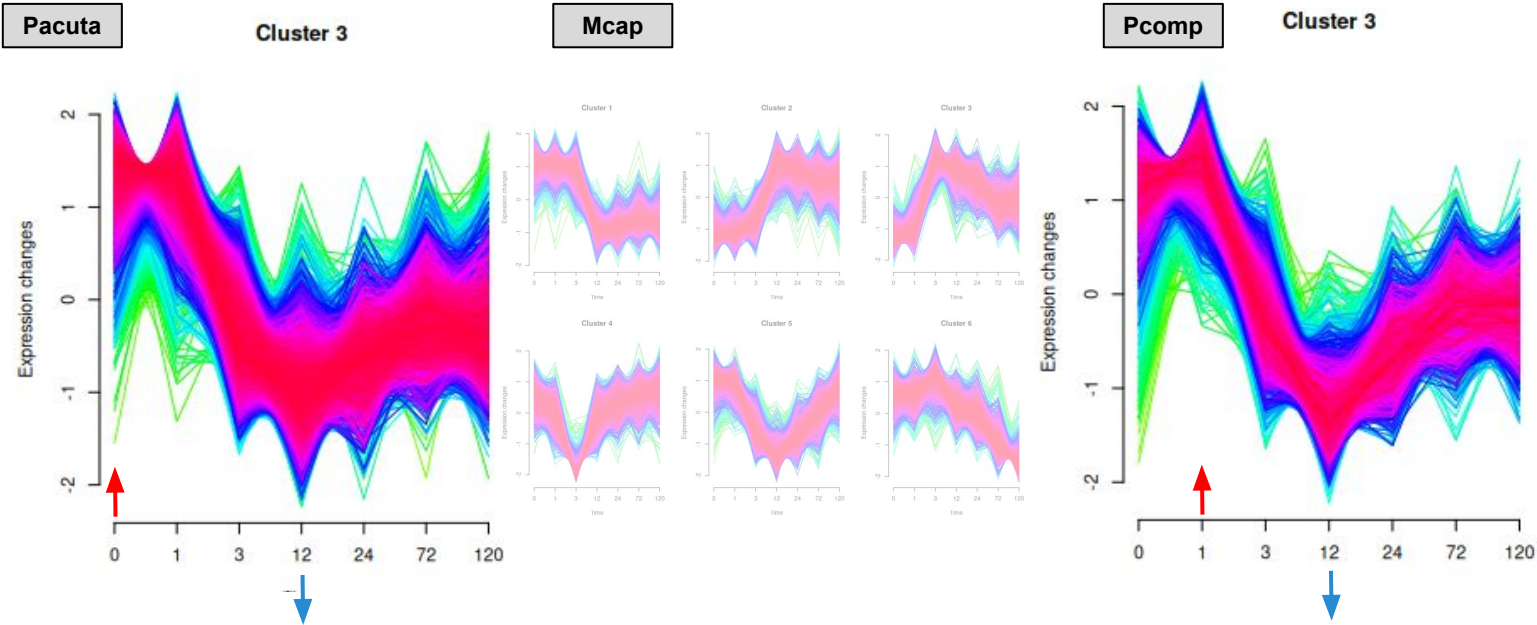
- Pacuta	- Mcap
- TGFa	- <i>VIT</i>
- Nrf2	- Pcomp
- NPC2	- <i>VIT</i>
- HSP70	- RhBG
- HSF1	- GOGAT
- GS	- <i>DNaj</i>
- GDH	- BI-1
- Foxo3	
- <i>DNaj</i>	
- AMT1	
- ST1C2	



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Early Dip (3h)	1	4	-	Pacuta & Mcap
U-shaped Dip (12h)	-	5	4	Mcap & Pcomp

Selected Heat Stress Genes (Manually curated):

- Pacuta
 - AMT1
 - ST1C2
 - RhBG
 - GOGAT
- Pcomp
 - HSF1
 - Glut8.SLC2A8
 - GDH
 - ALDH3B1



Color = Cluster Membership



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Early Dip (3h)	1	4	-	Pacuta & Mcap
U-shaped Dip (12h)	-	5	4	Mcap & Pcomp

Selected Heat Stress Genes (Manually curated):

- NID-2

ALOX12B
- Mcap

ST1C2

TGFa

RhBG

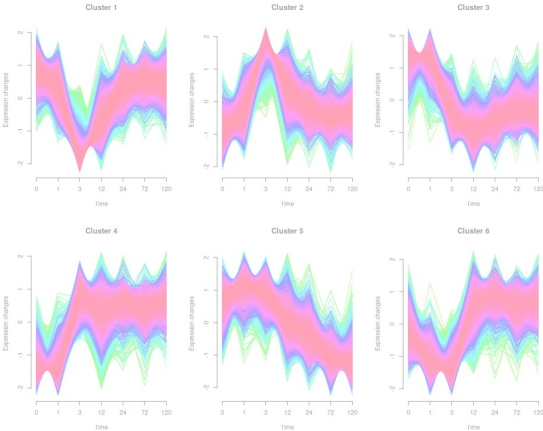
OGG1

GOGAT

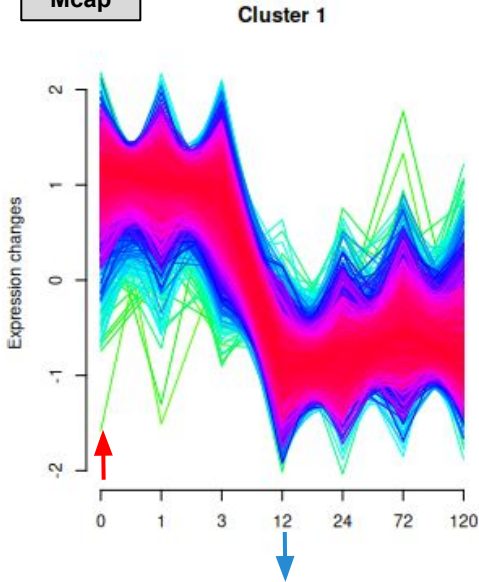
GDH
- Pcomp

NPC2

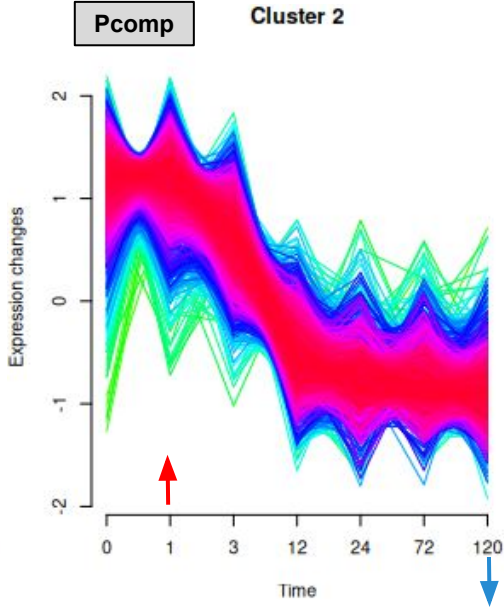
Pacuta



Mcap



Pcomp



Color = Cluster Membership



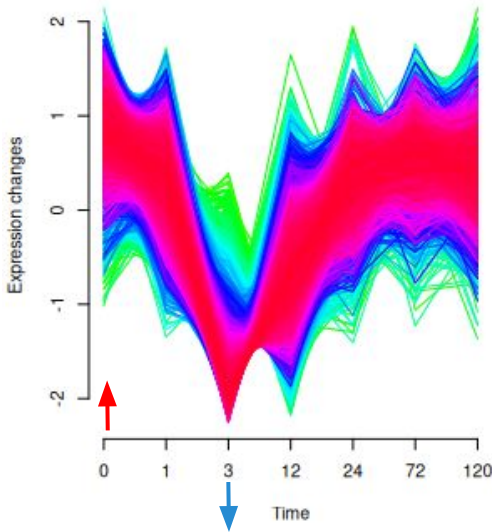
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Sustained Down (3h)	3	-	3	Pacuta & Pcomp
Sustained Down (12h)	-	1	2	Mcap & Pcomp
Early Dip (3h)	1	4	-	Pacuta & Mcap
U-shaped Dip (12h)	-	5	4	Mcap & Pcomp

Selected Heat Stress Genes (Manually curated):

- Pacuta
 - ST1C2
 - CAT
 - mTOR
 - ALDH3B1
- Mcap
 - GR
 - ST1C2

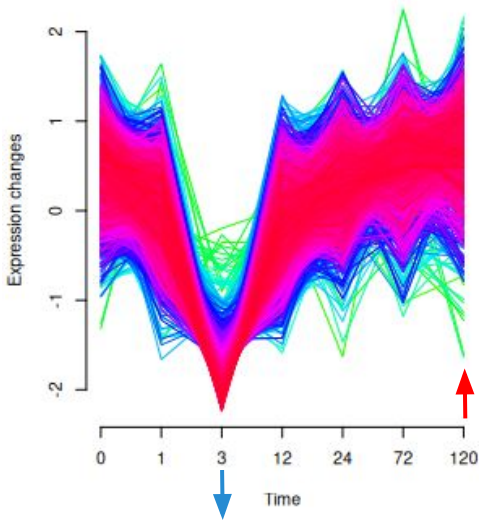
Pacuta

Cluster 1

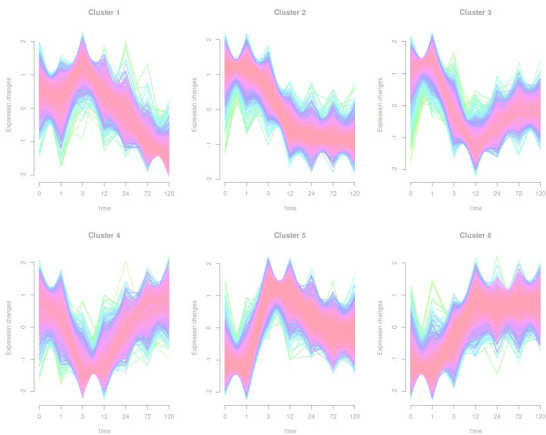


Mcap

Cluster 4



Pcomp



Color = Cluster Membership

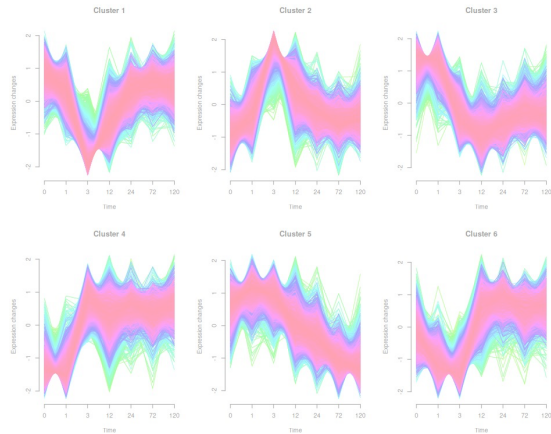


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Sustained Down (3h)	3	-	3	Pacuta & Pcomp
Sustained Down (12h)	-	1	2	Mcap & Pcomp
Early Dip (3h)	1	4	-	Pacuta & Mcap
U-shaped Dip (12h)	-	5	4	Mcap & Pcomp

Selected Heat Stress Genes (Manually curated):

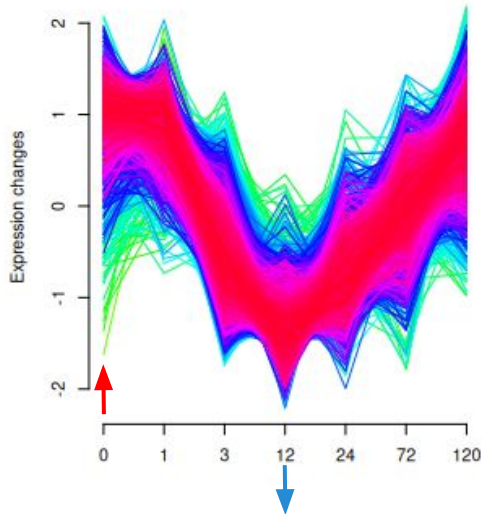
- Mcap
 - ST1C2
- Pcomp
 - ST1C2
 - AMT1

Pacuta



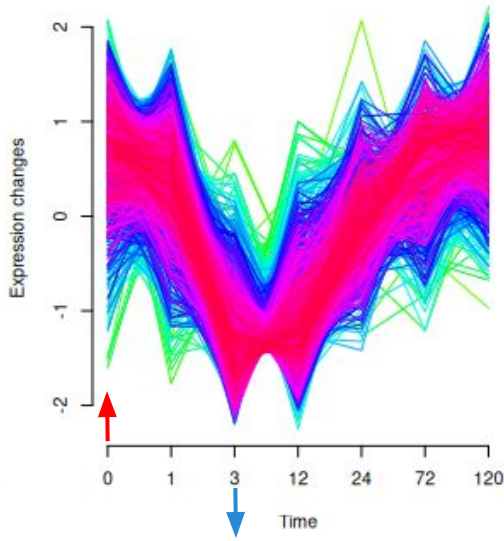
Mcap

Cluster 5



Pcomp

Cluster 4



Color = Cluster Membership

